Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df.shape
```

```
(11162, 17)
```

```
df.columns
```

```
Index(['age', 'job', 'marital', 'education', 'default', 'balance', 'housing',
       'loan', 'contact', 'day', 'month', 'duration', 'campaign', 'pdays',
       'previous', 'poutcome', 'deposit'],
      dtype='object')
```

Load Dataset

```
df = pd.read_csv("bank (2).csv")
df.head()
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previo
0	59	admin.	married	secondary	no	2343	yes	no	unknown	5	may	1042	1	-1	
1	56	admin.	married	secondary	no	45	no	no	unknown	5	may	1467	1	-1	
2	41	technician	married	secondary	no	1270	yes	no	unknown	5	may	1389	1	-1	
3	55	services	married	secondary	no	2476	yes	no	unknown	5	may	579	1	-1	
4	54	admin.	married	tertiary	no	184	no	no	unknown	5	may	673	2	-1	

Next steps: Generate code with df New interactive sheet

Data Cleaning

```
# Convert target variable
df['deposit'] = df['deposit'].map({'yes': 1, 'no': 0})

# Check missing values
print(df.isnull().sum())
```

```
age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays        0
previous     0
poutcome     0
deposit      0
dtype: int64
```

Funnel Definition

```
total = len(df)
leads = len(df[df['duration'] > 200])
customers = len(df[df['deposit'] == 1])

print("Contacted:", total)
```

```
print("Interested:", leads)
print("Converted:", customers)
```

```
Contacted: 11162
Interested: 6768
Converted: 5289
```

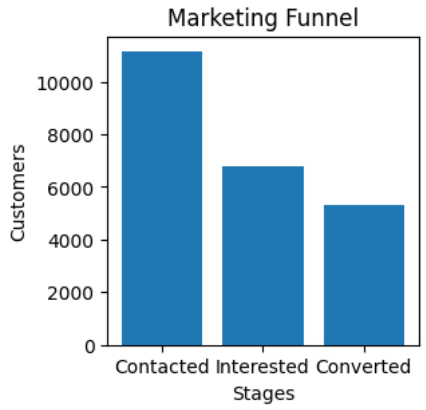
Conversion Rates

```
contact_to_lead = (leads / total) * 100
lead_to_customer = (customers / leads) * 100

print("Contact → Lead Conversion:", round(contact_to_lead, 2), "%")
print("Lead → Customer Conversion:", round(lead_to_customer, 2), "%")
```

```
Contact → Lead Conversion: 60.63 %
Lead → Customer Conversion: 78.15 %
```

Funnel Chart

```
stages = ['Contacted', 'Interested', 'Converted']
values = [total, leads, customers]

plt.figure(figsize=(3,3))
plt.bar(stages, values)
plt.title("Marketing Funnel")
plt.xlabel("Stages")
plt.ylabel("Customers")
plt.show()
```



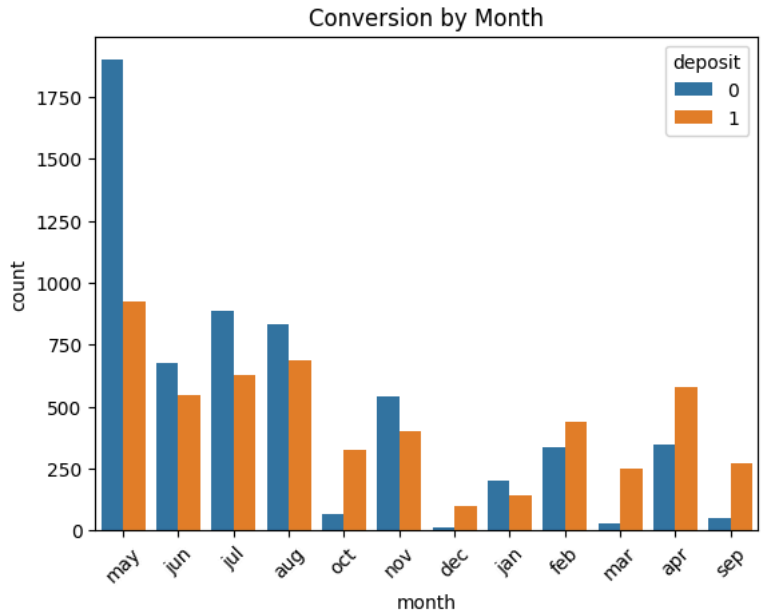
Drop-Off Analysis

```
drop1 = total - leads
drop2 = leads - customers

print("Drop-off (Contact → Lead):", drop1)
print("Drop-off (Lead → Customer):", drop2)
```

```
Drop-off (Contact → Lead): 4394
Drop-off (Lead → Customer): 1479
```

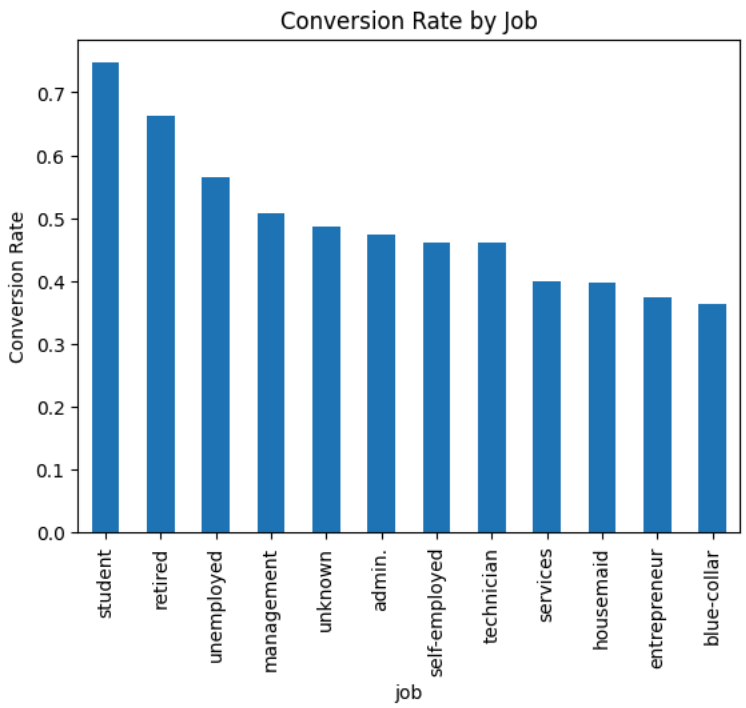
Conversion by Month

```
sns.countplot(x='month', hue='deposit', data=df)
plt.title("Conversion by Month")
plt.xticks(rotation=45)
plt.show()
```



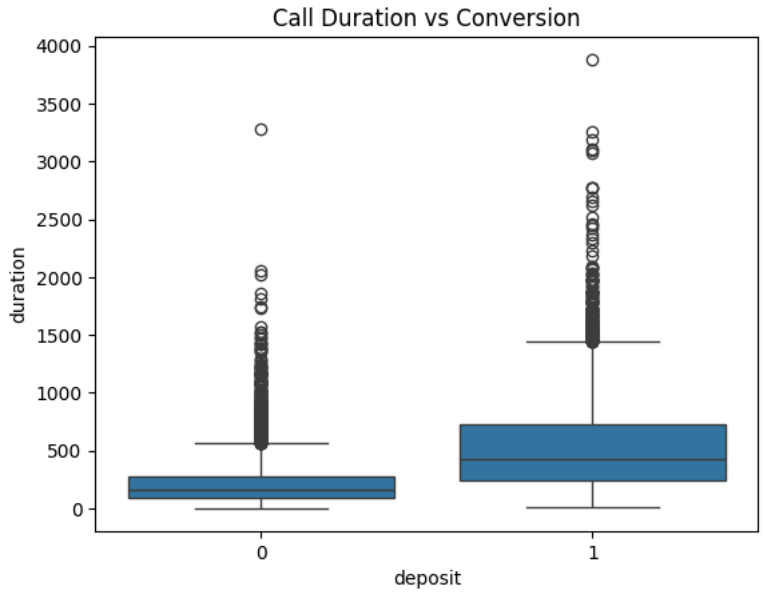
Conversion by Job

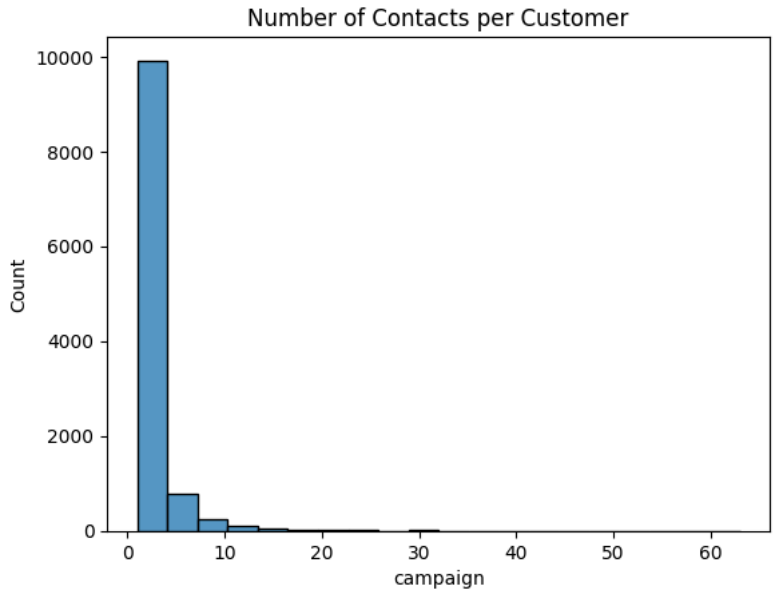
```
df.groupby('job')['deposit'].mean().sort_values(ascending=False).plot(kind='bar')
plt.title("Conversion Rate by Job")
plt.ylabel("Conversion Rate")
plt.show()
```



Call Duration Impact

```
plt.figure()
sns.boxplot(x='deposit', y='duration', data=df)
plt.title("Call Duration vs Conversion")
plt.show()
```



Campaign Effect

```
plt.figure()
sns.histplot(df['campaign'], bins=20)
plt.title("Number of Contacts per Customer")
plt.show()
```